SQL Aggregate Functions and operators

Exercise 2 : Istorhete Zwirhuys

Section 1: The student table

TABLE students

Student-id	name	age	department
1	Alice	20	IT
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	finance
5	Eve	22	HR

1. All distinct departments in Students

```
SELECT DISTINCT departments
FROM Students
```

Results table →

department
IT
HR
Finance

2. Average age of Students per department

```
SELECT department,
       AVG(age) AS average-age-per-dep
FROM student
GROUP BY department
```

output table

department	avg_age
IT	20,5
HR	22
Finance	23

03 Departments with more than one student

```
SELECT department,
    COUNT (*) AS Student_count per_department
FROM Student
GROUP BY department
HAVING COUNT (*) > 1;
```

Results table

department	Student_count
IT	2
HR	2

04 Get all students whose age is between 21 and 23

```
SELECT Student_id, name, age
FROM Students
WHERE age is BETWEEN 21 AND 23;
```

Student_id	name	age
2	Bob	22
3	Charlie	21
4	Diana	23
5	Eve	22

05 List all students in IT or HR department who are older than 21:

```
SELECT   student_id, name, age, department
FROM     students
WHERE    department IN ('IT', 'HR')
AND      age > 21
```

Results

student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

Section 2: The courses Table

06 Total credits per department, for department with more than 5 total credits

```
                department's
SELECT   sum(credits) As total_credits
 FROM    courses
 GROUP BY  departments
Having Sum(credits) > 5;
```

TABLE courses

course_id	course_name	department	credits
101	SQL Basics	IT	3
102	Python	IT	4
103	Data Science	IT	4
104	Excel	finance	2
105	Statistics	HR	3

06 Total credits per department, for department (>5 total credits)

~~SELECT~~

```
SELECT   department,
         Sum (credits) AS   total_credits
FROM     courses
GROUP BY department
HAVING   sum (credits) > 5;
```

IT = 3+4+4
= 11

Finance = 2 HR = 3

∴

Results table

Department	total - credits
IT	11

07 List all courses that do not have ~~courses~~ 4 credits

```
SELECT course_id, course_name, department, credits
FROM   courses
WHERE NOT credits = 4;
```

Course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

08) Top 3 courses by credits in descending order.

```
SELECT course_id, course_name, credits
FROM   courses
ORDER BY credits DESC
LIMIT  3;
```

Course_id	course_name	credits
102	Python	4
103	Data science	4
101	SQL Basics	3

Section 3: The enrolment table

TABLE enrollments

enrollment_id	student_id	course_id	grade
1	1	101	85
2	2	102	78
3	3	103	90
4	4	104	88
5	5	105	82

09) Get max, min, and average grade across all enrollment

```
SELECT  MAX(grade) AS max_grade_enrollment,
        MIN(grade) AS min_grade_enrollment,
        Avg(grade) AS avg_grade_enrollment,
FROM    enrollments;
```

Results table

max_grade	min_grade	avg_grade
90	78	$\frac{(85+78+90+88+82)}{5}$ = 84.6

10. Count how many enrollment exist per course

```
SELECT course_id
       ~~COUNT (*)~~
  COUNT (*) AS enrollment_count_per course
FROM enrollments
GROUP BY course_id
```

Results table (output)

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

SECTION 4 : The salaries table

~~OR~~ TABLE salaries

employee_id	name	department	salary	bonus
1	Tom	IT	60000	5000
2	Jerry	HR	55 000	4000
3	Spike	finance	70 000	6000
4	Tyke	IT	62 000	5500
5	Butch	HR	54 000	3500

11- find total salary and total bonus per department

```
SELECT   department,
         Sum(salary) As total_salary,
         Sum (bonus) As total_bonus
FROM     salaries
GROUP BY department
```

Output

department	total_salary	total_bonus
IT	122 000	10500
finance	70 000	6000
HR	109 000	7500

12- department where avarage salary is above 55,000

```
SELECT   department,
         AVG (salary) As avg_salary
From     salaries
GROUP BY department
HAVING   AVG > 55000
```

$IT = \frac{122\,000}{2} = 61\,000$

finance = 70 000

$HR = \frac{109\,000}{2} = 53\,500$

department	avg_salary
IT	61 000
~~HR~~	
finance	53 500

(13) List all employees whose salary plus bonus is greater than 60 000

~~SELECT employees id, name, department, salary + bonus~~

```
SELECT employees id,
       name, department,
       salary, bonus
       AS total_salary_bonus_>60000
FROM salaries
WHERE salary AND bonus >60000
```

```
SELECT employees id,
       name, salary,
       bonus,
       Salary + bonus AS total_remuneration
FROM salaries
WHERE Salaries + bonus > 60 000;
```

Tom	60000 + 5000	65000 ✓
Jerry	55000+4000	59000
Spike	70000+6000	76000
Tyke	62000 + 5500	67500 ✓
Butch	54000+3500	57500

Output table

employee_id name Salary bonus total

employee_id	name	Salary	bonus	total
~~T~~ 1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

Section 5 The project table

TABLE projects

Project_id	project_name	department	budget
1	AI App	IT	120 000
2	Payroll system	finance	80 000
3	Dashboard	IT	150 000
4	Website	Marketing	60 000
5	HR Portal	HR	50 000

4) Show total and average per department, Only departments with avarage budget > 70 000

```
SELECT department,
       sum(budget) AS total_budget,
       AVG (budget) AS avg_budget
FROM projects
GROUP By department
HAVING AVG (budget)> 70000 ;
```

department	total_budget	avg_budget
IT	270000	135 000
finance	80 000	80 000

15) All projects with budgets between 50 000 and 120 000, excluding Marketing department

```
SELECT project_id, project name,
       department, budget
FROM projects
WHERE budget BETWEEN 50 000 AND 120 000
AND department <> 'Marketing';
```

project_id	project_name	department	budget
1	AI App	IT	120 000
2	Payroll system	finance	80 000
3	HR portal	HR	50 000